Light

GCSE Computer Science - Programming Testing and Robustness (Set 3)

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GCSE • OCR J277 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (OCR J277)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

What does a presence check verify?

- ○ **A.** That data is within a specified numerical range
- ○ **B.** That data has been entered and the field is not left empty
- ○ **C.** That data matches a specific pattern or format
- ○ **D.** That data is of the correct data type

Q2 (1 mark)

What does a length check verify?

- ○ **A.** That data falls within minimum and maximum numerical values
- ○ **B.** That the data contains specific required characters
- ○ **C.** That the number of characters entered is within acceptable limits
- ○ **D.** That the data type matches what is expected

Q3 (1 mark)

What does a type check verify?

- ○ **A.** That the input data is of the expected data type, such as integer, string, or boolean
- ○ **B.** That the input contains a specific number of characters
- ○ **C.** That the input is not empty
- ○ **D.** That the input falls within a specified range of values

Q4 (1 mark)

What is the purpose of a trace table in programming?

- ○ **A.** To record which users have accessed the program
- ○ **B.** To track how variable values change as a program executes line by line
- ○ **C.** To list all the functions used in a program

- ○ **D.** To measure how fast a program runs

Q5 (1 mark)

Which of the following would cause a syntax error in Python?

- ○ **A.** Using the wrong formula in a calculation
- ○ **B.** Dividing a number by zero
- ○ **C.** Forgetting to close a string with a quotation mark
- ○ **D.** Using a variable before assigning a value to it

Q6 (1 mark)

Which of the following describes a logic error?

- ○ **A.** A loop that runs one too many times, causing incorrect output
- ○ **B.** A misspelled keyword that the interpreter cannot recognise
- ○ **C.** A missing bracket that prevents the code from running
- ○ **D.** A variable name that contains an illegal character

Q7 (1 mark)

Why should programmers use named constants instead of 'magic numbers' in their code?

- ○ **A.** Constants give meaningful names to values, making code easier

to understand and update

- ○ **B.** Constants use less memory than regular numbers
- ○ **C.** Constants make the program run faster
- ○ **D.** Constants automatically update when requirements change

Q8 (1 mark)

Why is using local variables generally preferred over global variables?

- ○ **A.** Local variables do not need to be declared before use
- ○ **B.** Local variables can store larger values than global variables
- ○ **C.** Local variables are automatically backed up by the system
- ○ **D.** Local variables are only accessible within their sub program, reducing the risk of unintended changes

Q9 (1 mark)

Which IDE feature helps programmers identify exactly where their code stops working?

- ○ **A.** Syntax highlighting
- ○ **B.** Code auto-completion
- ○ **C.** Breakpoints and step-through debugging
- ○ **D.** File management

Q10 (1 mark)

What is input sanitisation?

- ○ **A.** Removing or escaping potentially dangerous characters from user input before processing
- ○ **B.** Encrypting user input for secure storage
- ○ **C.** Converting input to uppercase for consistency
- ○ **D.** Checking that all required fields have been completed

Q11 (1 mark)

Why should error messages be clear and helpful to users?

- ○ **A.** To make the program run faster
- ○ **B.** To help users understand what went wrong and how to fix it
- ○ **C.** To reduce the file size of the program
- ○ **D.** To hide technical details from hackers

Q12 (1 mark)

Why should test data include values that the program should reject?

- ○ **A.** To check the program crashes gracefully when given bad data
- ○ **B.** To test the program's speed with different inputs
- ○ **C.** To make the testing process take longer

○ **D.** To verify the validation correctly identifies and rejects invalid or erroneous input

Q13 (2 marks)

A registration form requires users to enter their name.
The programmer wants to ensure users do not leave this field blank.
Which validation check should be applied?

- ○ **A.** Range check
- ○ **B.** Format check
- ○ **C.** Presence check
- ○ **D.** Type check

Q14 (2 marks)

A system requires PINs to be exactly 4 digits long.
Which validation check would verify this requirement?

- ○ **A.** Length check to ensure exactly 4 characters are entered
- ○ **B.** Range check to ensure the number is between 0000 and 9999
- ○ **C.** Presence check to ensure something was entered
- ○ **D.** Format check to ensure it contains letters

Q15 (2 marks)

A programmer uses a trace table to debug the following code:

```
x = 5
y = 3
```

```
x = x + y
y = x - y
x = x - y
```

What are the final values of x and y?

- ○ **A.** x = 5, y = 3
- ○ **B.** x = 8, y = 5
- ○ **C.** x = 3, y = 5
- ○ **D.** x = 3, y = 8

Q16 (2 marks)

A student writes the following Python code:

```
count = 0
```

```
while count < 5
    print(count)
    count = count + 1
```

What error is present in this code?

- ○ **A.** Logic error - the loop will never end
- ○ **B.** Syntax error - missing colon after the while condition
- ○ **C.** Runtime error - count is not initialised
- ○ **D.** Logic error - count should start at 1 not 0

Q17 (2 marks)

A program should display 'Pass' for scores of 50 or above.
The code is: `if score > 50:`
A student with exactly 50 marks is shown 'Fail'.

What type of error is this and what is wrong?

- ○ **A.** Logic error - using > instead of >= means 50 is excluded from passing
- ○ **B.** Syntax error - the colon should not be there
- ○ **C.** Runtime error - score was not defined
- ○ **D.** No error - 50 should result in Fail

Q18 (2 marks)

A weather application accepts temperature readings from -50 to 60 degrees Celsius.
Which set contains appropriate boundary test data?

- ○ **A.** 0 and 25
- ○ **B.** Only 0
- ○ **C.** -100 and 100
- ○ **D.** -50, -49, 59, and 60

Q19 (2 marks)

A system requires usernames to be 3-12 characters, containing only letters and numbers.
Which input should be rejected?

- ○ **A.** ab
- ○ **B.** user123
- ○ **C.** TestUser

- ○ **D.** Player1

Q20 (2 marks)

A program divides two numbers entered by the user.
What defensive design technique should prevent the program crashing if the user enters 0 as the divisor?

- ○ **A.** Checking if the divisor is zero before performing the division
- ○ **B.** Using comments to explain the code
- ○ **C.** Using meaningful variable names
- ○ **D.** Adding indentation to the code

Q21 (2 marks)

A UK phone number field requires input in the format '07XXX XXXXXX' where X is a digit.
Which validation check verifies this pattern?

- ○ **A.** Presence check
- ○ **B.** Range check
- ○ **C.** Length check
- ○ **D.** Format check

Q22 (2 marks)

After fixing a bug, a programmer re-runs all previous tests to ensure the fix has not broken other parts of the program.
What is this practice called?

- ○ **A.** Iterative testing
- ○ **B.** Final testing
- ○ **C.** Regression testing
- ○ **D.** Acceptance testing

Q23 (2 marks)

A programmer has written a 200-line function that validates user input, processes data, generates a report, and sends an email.
How should this be improved for maintainability?

- ○ **A.** Add more comments to explain each section
- ○ **B.** Remove all blank lines to make it shorter
- ○ **C.** Rename the function to describe all its tasks
- ○ **D.** Split it into separate sub programs, each handling one specific task

Q24 (3 marks)

An online shop must verify customers are over 18 to purchase certain products.
Analyse which combination of validation checks would provide robust age verification from a date of birth input.

- ○ **A.** Presence check only
- ○ **B.** Presence check, format check (valid date pattern), range check (date not in future), and calculation to verify resulting age is 18 or over
- ○ **C.** Just asking users to confirm they are over 18

○ **D.** Length check to ensure the date has enough characters

Q25 (3 marks)

A program calculates employee bonuses. Some employees receive incorrect amounts, but the program runs without errors.
Analyse the most effective systematic approach to identify and fix this problem.

○ **A.** Use a trace table or debugger to step through the calculation with known input values, comparing actual variable values at each step against expected values to identify where the calculation diverges from the specification

○ **B.** Delete the code and start again from scratch

○ **C.** Add more comments to the code

○ **D.** Run the program many times hoping the error will reveal itself

Q26 (3 marks)

A web application displays user comments on a page without sanitising the input first.
Analyse the security risk this creates and how it should be addressed.

○ **A.** This creates a Cross-Site Scripting (XSS) vulnerability where attackers can inject malicious scripts that execute in other users' browsers; user input should be sanitised by escaping or removing HTML and JavaScript before display

○ **B.** There is no security risk as long as the database is password protected

○ **C.** The only risk is that comments might contain spelling mistakes

○ **D.** This is only a problem if the server has slow internet connection

Q27 (3 marks)

A school develops a new student registration system that handles personal data, generates timetables, and records exam entries. Analyse which testing strategy would be most appropriate.

○ **A.** Test only with one student record since all records work the same way

○ **B.** Skip testing because the programmers are experienced

○ **C.** Comprehensive testing using all test data types for each module during development (iterative), integration testing when modules are combined, and final system testing against requirements; include boundary cases for dates and grades, invalid data to test validation, and erroneous data types to test error handling

○ **D.** Only test after complaints are received from users

Q28 (3 marks)

A programmer inherits code containing these issues: the value 1.175 appears in 12 different places representing a tax multiplier; variable names include 'a', 'temp', 'x2'; and identical calculation code is copied in 8 different locations.
Analyse how these issues affect maintainability and what improvements are needed.

○ **A.** The code is fine because it currently works correctly

○ **B.** Repeated magic numbers make tax rate changes error-prone (must find all 12 occurrences); meaningless names make understanding the code difficult; duplicated code means bugs must be fixed 8 times; improvements: use a named constant for the tax rate, rename variables descriptively, and extract the repeated calculation into a reusable sub program

- ○ **C.** Only the variable names need changing
- ○ **D.** The code should be deleted without attempting to understand it

Q29 (3 marks)

A program reads data from a file, processes it, and writes results to another file.
Analyse what defensive design measures should be implemented to make this program robust.

- ○ **A.** No defensive measures are needed for file operations
- ○ **B.** Check the input file exists before reading; handle file access errors gracefully; validate data read from the file; check write permissions before attempting output; handle disk full scenarios; provide meaningful error messages for each failure type rather than crashing
- ○ **C.** Only check that the files have the correct names
- ○ **D.** Run the program twice to ensure it works

Q30 (3 marks)

A program accepts percentage grades from 0 to 100 and displays 'Distinction' (70-100), 'Merit' (55-69), 'Pass' (40-54), or 'Fail' (0-39).
Analyse which test plan would provide the most thorough verification.

- ○ **A.** Test only with 50 since it is in the middle
- ○ **B.** Test only with 100 since it is the highest grade
- ○ **C.** Test with any three random numbers
- ○ **D.** Test with 0, 39, 40, 54, 55, 69, 70, 100 (all grade boundaries), plus -1 and 101 (invalid), plus 'fifty' (erroneous); verify each produces the

correct category or appropriate rejection

Total: **55** marks • Time: **30 min**

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